

WASP-12b

R-1688

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190227 · created 2026-08-02T12:46:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458541.77205 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.044
Transit depth (Rp/Rs)^2	7.4e-05 +/- 0.0075 [%]
Orbital Inclination (inc)	83.4 +/- 2.9
Airmass coefficient 1 (a1)	0.86 +/- 0.29
Airmass coefficient 2 (a2)	0.061 +/- 0.062
Scatter in the residuals of the lightcurve fit is	34.09 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	6.0
Transit Duration (day)	0.1124 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.044 vs 0.1178 (deviation 2.32σ)
Duration fit vs published	0.1124 d vs 0.125 d (deviation 1.43σ)
Mid-time O-C	-0.8 min
Depth SNR	0.0
Residual scatter	34.09 %

Light curve

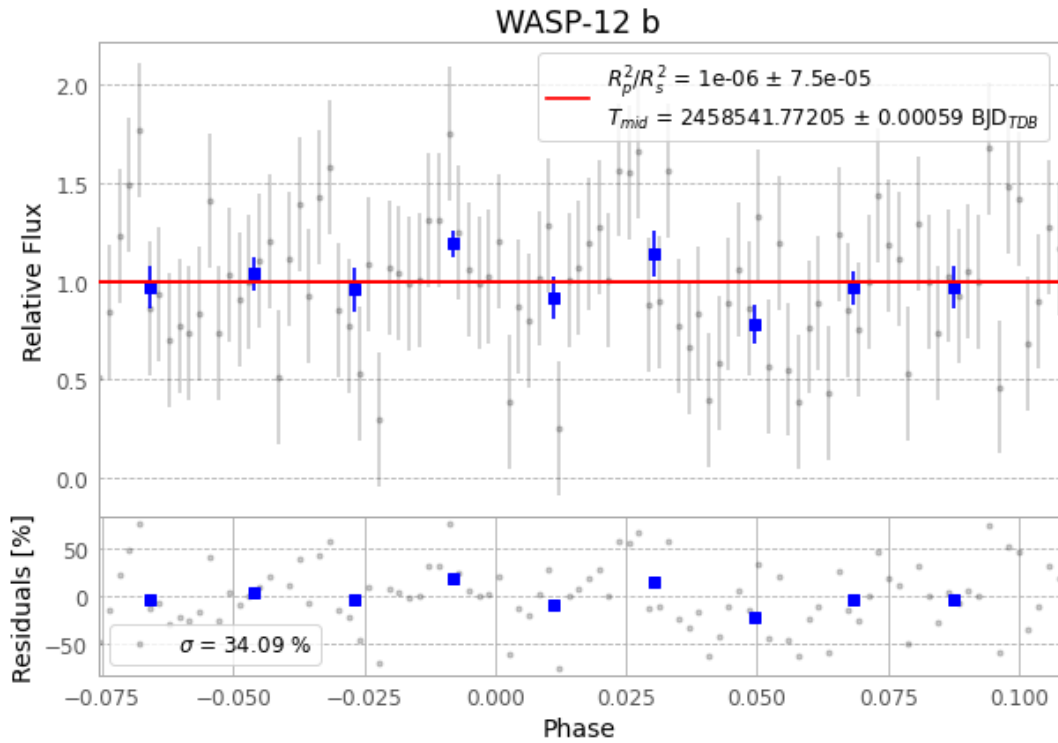


Figure 1 — FinalLightCurve_WASP-12 b_2019-02-26.png (SHA-256 3ad893cd4eda43c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [595, 182], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d758907bbb7369ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), WASP-12190227042712.FITS → WASP-12190227091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190227.log (1b4ab785c6a2...), FinalLightCurve_WASP-12 b_2019-02-26.png (3ad893cd4eda...), FinalLightCurve_WASP-12 b_2019-02-26.csv (f89d639495b0...)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)